

PRINCIPAL COMPONENT REGRESSION, RIDGE REGRESSION AND RIDGE PRINCIPAL COMPONENT REGRESSION IN SPECTROSCOPY CALIBRATION

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SUMMARY

Ridge regression (RR) and principal component regression (PCR) are two popular methods intended to overcome the problem of multicollinearity which arises with spectral data. The present study compares the performances of RR and PCR in addition to ordinary least squares (OLS) and partial least squares (PLS) on the basis of two data sets. An alternative procedure that combines both PCR and RR is also introduced and is shown to perform well. Furthermore, the performance of the combination of RR and PCR is stable in so far as sufficient information is taken into account. This result suggests discarding those components that are unquestionably identified as noise, when the ridge constant tackles the degeneracy caused by components with small variances. © 1997 by John Wiley & Sons, Ltd.

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INTRODUCTION

Near-infrared (NIR) spectroscopy is widely used for rapid determination of the chemical composition of food and feed products. This firstly involves building calibration equations between the spectral data and the chemical components of interest. For this purpose, various statistical tools are available. The most popular is undoubtedly the linear regression model, which may be stated as

$$\mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\epsilon} \quad (1)$$

where $\mathbf{X}$ is the $n \times p$ matrix of spectral data, $\mathbf{y}$ is the n -vector of chemical responses, $\boldsymbol{\beta}$ is the p -vector of regression coefficients and $\boldsymbol{\epsilon}$ is the n -vector of independent and normally distributed errors.

In the case of continuous signals such as NIR spectra the number p of predictors in the matrix $\mathbf{X}$ often exceeds the number n of observations. In addition, there is a high degree of near collinearity among the predictors. It is common knowledge that under these conditions, ordinary least squares (OLS) estimations of regression coefficients may be very unstable, leading to very poor prediction accuracy.¹

Alternative methods to OLS are proposed in order to overcome this problem. The most popular in chemometrics are principal component regression (PCR) and partial least squares (PLS).² Ridge regression (RR) is another alternative to OLS regression in the case of the existence of near collinearity among predictors.³ Different authors have recently studied the interest of these respective

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methods for multivariate calibration in chemometrics.⁴⁻⁶

The rationale behind PCR, PLS and RR is to reduce the variability of the β estimations by shrinking the coefficient vector away from the OLS solution.^{4,7} PCR and PLS are discrete shrinkage methods because the parameter of interest is the number of latent variables introduced in the model, whereas RR is a continuous shrinkage method as it depends on a scalar parameter k . In PCR the latent variables are chosen among the principal components of $\mathbf{X}$ and in PLS the latent variables included in the model are iteratively computed using information from both $\mathbf{X}$ and $\mathbf{y}$.² The optimal number of latent variables in PCR or PLS is usually determined by a cross-validation technique. Several techniques for the determination of an optimal ridge parameter k , including procedures based on cross-validation, have been discussed by various authors.⁸⁻¹¹

The present study focuses on the use of PCR and RR for NIR calibration with numerical comparisons of the PLS method. A common framework was set up for introducing PCR and RR. A new alternative method that combines these two shrinking methods was also suggested. For the determination of the optimal number of components to be included in the PCR model, or for the determination of the ridge parameter k , the PRESS statistic was computed according to two techniques: the LOO (leave-one-out) technique or by using a formula.^{11,12} This last approach may be preferred as it is less time-consuming than the LOO technique. Two data sets were involved in the comparison of the methods. The first one was a published data set¹³ with $n > p$. The second one was a set of NIR spectra collected in our laboratory in order to predict the technological behaviour of mixtures to be pelleted in the feedstuff industry. In this second case $n \ll p$.

METHODS

Principal component regression and ridge regression

By way of introducing simply PCR and RR, the matrix $\mathbf{X}$ of predictors (assumed to be centred) is expressed in terms of its p principal components. For this purpose, $\mathbf{D}$ is defined as the $p \times p$ diagonal matrix of the eigenvalues of $\mathbf{X}^T \mathbf{X}$ and $\mathbf{P}$ is the matrix of normalized eigenvectors. Then

$$\mathbf{X}^T \mathbf{X} = \mathbf{P} \mathbf{D} \mathbf{P}^T \quad (2)$$

with $\mathbf{P}^T \mathbf{P} = \mathbf{P} \mathbf{P}^T = \mathbf{I}$ ($\mathbf{I}$ being the identity matrix). The linear regression model (equation (1)) may be expressed as

$$\mathbf{y} = \mathbf{X} \mathbf{P} \mathbf{P}^T \boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (3)$$

or

$$\mathbf{y} = \mathbf{Z} \boldsymbol{\alpha} + \boldsymbol{\varepsilon} \quad (4)$$

where $\mathbf{Z} = \mathbf{X} \mathbf{P}$ (matrix of principal components) and $\boldsymbol{\alpha} = \mathbf{P}^T \boldsymbol{\beta}$.

If $n \leq p$, the eigenvalues λ_j of $\mathbf{X}^T \mathbf{X}$ are equal to zero for $j = n$ to p . As a rule, $q = \min(n - 1, p)$ principal components are computable and the OLS estimation of the vector $\boldsymbol{\alpha}$ of coefficients is

$$\mathbf{a}(0, q) = \mathbf{D}^{-1} \mathbf{Z}^T \mathbf{y} \quad (5)$$

where

$$\mathbf{D}^{-1} = \text{diag} \left(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_q} \right) \quad (6)$$

As the spectral variables deviate further from orthogonality, the last eigenvalues may be very close to zero. As a result, their inverses may be very large, leading to several undesirable consequences: coefficient estimates far removed from true coefficients, poor accuracy of prediction, etc.

In PCR the estimation of α is achieved by retaining only the first c principal components:

$$\mathbf{a}(0, c) = \mathbf{D}_{0,c}^{-1} \mathbf{Z}^T \mathbf{y} \quad (7)$$

where

$$\mathbf{D}_{0,c}^{-1} = \text{diag} \left(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_c}, 0, \dots, 0 \right) \quad (8)$$

In RR a constant k is added to all eigenvalues, leading to the estimation:

$$\mathbf{a}(k, q) = \mathbf{D}_{k,q}^{-1} \mathbf{Z}^T \mathbf{y} \quad (9)$$

where

$$\mathbf{D}_{k,q}^{-1} = \text{diag} \left(\frac{1}{\lambda_1 + k}, \frac{1}{\lambda_2 + k}, \dots, \frac{1}{\lambda_q + k} \right) \quad (10)$$

It appears from equations (7)–(10) that PCR and RR shrink the coefficients associated with the principal components $\mathbf{Z}$ in different ways. In PCR, principal components associated with eigenvalues $\lambda_{c+1}, \dots, \lambda_q$ are discarded in order to eliminate the noise factor from the data. The limitations of this procedure are that the principal components are retained or discarded on the basis of the calibration set and it may happen that the removed principal components contain useful information for purposes of prediction. On the other hand, if principal components with small variances (small eigenvalues) are introduced into the model, the variances of coefficient estimates are inflated. In RR, all eigenvalues are modified by adding a constant k . In general, k is small and has no perceptible impact on the largest eigenvalues and a significant impact on the smallest ones. All principal components are retained, but the degeneracy caused by the latter is tackled by the introduction of the parameter k .

Furthermore, equations (7)–(10) prompt a new estimation procedure for the vector α of coefficients:

$$\mathbf{a}(k, c) = \mathbf{D}_{k,c}^{-1} \mathbf{Z}^T \mathbf{y} \quad (11)$$

where

$$\mathbf{D}_{k,c}^{-1} = \text{diag} \left(\frac{1}{\lambda_1 + k}, \frac{1}{\lambda_2 + k}, \dots, \frac{1}{\lambda_c + k}, 0, \dots, 0 \right) \quad (12)$$

The idea is to discard the principal components unquestionably identified as noise and to add a constant in order to handle the problem caused by the remaining small eigenvalues. Such a procedure combines the principles of both PCR and RR and will be referred to as ridge principal component regression (RPCR).

Determination of parameters k and c

The number c^* of principal components included in the prediction equation is usually chosen by minimizing the PRESS statistic

$$\text{PRESS} = \sum_{i=1}^n (y_i - \hat{y}_{li})^2 \quad (13)$$

where $\hat{y}_{li}$ is the i th value predicted from the model fitted with a reduced calibration set obtained by removing the i th observation. The strategy is to use a LOO (leave-one-out) technique which requires fitting n different regressions for each value of c . However, it is possible to calculate PRESS from the results of a single least squares fit.¹² With $e_i = y_i - \hat{y}_i$ the ordinary residual for the i th observation, using c principal components, the i th PRESS residual is given by

$$y_i - \hat{y}_{li} = \frac{e_i}{1 - h_{ii}} \quad (14)$$

where h_{ii} is a diagonal element of the matrix $\mathbf{Z} \mathbf{D}_{0,c}^{-1} \mathbf{Z}^T$.

Thus the PRESS statistic reduces to

$$\text{PRESS}_{\text{formula}} = \sum_{i=1}^n \left(\frac{e_i}{1 - h_{ii}} \right)^2 \quad (15)$$

Nevertheless, it must be noted that in the case of an ill-conditioned design (high collinearity among predictors), certain discrepancies arise between the results provided by the formula and those obtained by effectively performing a LOO technique.

The same cross-validation strategy can be applied to determine an optimal value k^* for the ridge parameter. Instead of using a LOO technique, the cross-validated residuals can easily be calculated according to equation (14), h_{ii} being now a diagonal element of the matrix $\mathbf{Z} \mathbf{D}_{k,q}^{-1} \mathbf{Z}^T$. This extension is valid because the ridge estimator is a linear estimator.¹¹

Similarly, when RR is performed on a subset of c principal components (RPCR), the parameter k may be determined as a minimizer of $\text{PRESS}_{\text{formula}}$ (equation (15)) with h_{ii} defined as a diagonal element of the matrix $\mathbf{Z} \mathbf{D}_{k,c}^{-1} \mathbf{Z}^T$.

All computations were performed using SAS/IML language.¹⁴

DATA

Two data sets relating to calibration problems in spectroscopy were considered to evaluate the predictive ability of RR, PCR and PLS and the performance of RPCR.

Data set 1

The first data set is delivered with the Unscrambler software¹³ and is used as a tutorial for multivariate regression with PLS. The problem was to determine the concentration of a dye (cytochrome C) in water solutions. The concentration determination was based on the light absorbance in the visible

range. $p=16$ different wavelength channels were selected as predicting variables and $n=27$ samples of solution were prepared as a calibration set (one observation identified as an outlier has been removed). The dependent variable, recorded on the calibration set, had an average of 3.74 ml and a standard deviation of 2.82 ml.

Data set 2

In the second example the aim was to predict the durability of pellets in the feedstuff industry from NIR spectra of the mixtures to be pelleted. Feedstuffs are often obtained by mixing raw materials and then pelleting. The technological qualities of pellets can be measured by their durability, which is defined as the proportion of pellets that have not been broken during shaking in standardized conditions. The durability varies with the raw materials included in the pellets and with their particle size and biochemical composition. It is important to assess the durability of the mixture before pelleting. NIR spectroscopy may be used for this purpose as NIR spectra depend on both the particle size and biochemical composition. In the present work a set of $n=99$ mixtures of eleven usual raw materials was studied. These raw materials were maize, wheat, soya, cassava, bran, pea, corn gluten feed, rapeseed meal, barley and two liquids: oil and molasses. The proportions in the mixtures were computed according to an experimental design taking into account technological and nutritional constraints. The details of the experimental design are given in Reference 15. The mixtures were pelleted in the experimental press of the INRA Center, Nantes. The durability of the pellets was measured in the Holmen Pellet Tester (Holmen Chemicals Ltd., PO Box 2, Basing View, Basingstoke, Hants, U.K.). The average durability of the 99 samples was 76.99% with a standard deviation of 19.67%.

The NIR spectra of the 99 mixtures were recorded before pelleting on an infraAlyzer 500 (Technicon) spectrometer between 1100 and 2500 nm in steps of 4 nm. A total of $p=351$ spectral variables were available. Figure 1 shows a subset of these spectra.

As a large collection of 99 samples was available, the total data set was split into a calibration set and a validation set. The calibration set was formed by selecting $n=50$ samples at random from the whole set of samples and the validation set was composed of the $n'=49$ remaining samples. The model was fitted using the calibration samples, while the validation samples gave an estimation of the prediction error.

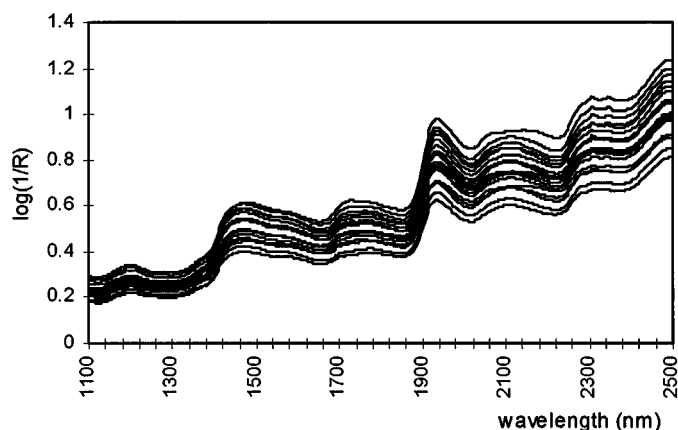


Figure 1. Near-infrared spectra of mixtures of raw feedstuffs

Data treatment

PCR, RR and RPCR were applied on the two data sets. For numerical comparison, PLS was also performed using the Unscrambler software.¹³ In addition, OLS results are reported in order to compare the improvement of the four alternative methods.

For the first data set the optimal values of the models parameters, i.e. c^* for PCR, k^* for RR, (c^*, k^*) for RPCR and the number of latent variables (a^*) in PLS, were determined in order to minimize the ordinary PRESS criterion. As a matter of fact, n and p were sufficiently small to allow computations with the LOO strategy.

In the case of the second data set the determination of the optimal value for the biased regression methods was carried out either by the LOO cross-validation technique or by the optimization of the PRESS formula (equation (15)) as described earlier. As the time required with the LOO technique was very long, the formula was used for RPCR and RR. For PCR the LOO technique was used because $\text{PRESS}_{\text{formula}}$ led to unreliable results. The optimal number of latent variables in PLS was determined using a LOO strategy.

For the optimal model of each method, as for the OLS method, the prediction ability was estimated by the cross-validation error:

$$\text{CV} = \frac{\text{PRESS}}{n} \quad (16)$$

When a validation set was available (data set 2), errors of prediction were also evaluated by the mean square error of prediction:

$$\text{MSEP} = \frac{1}{n'} \sum_{j=1}^{n'} (y_j - \hat{y}_j)^2 \quad (17)$$

where $\hat{y}_j$ is the prediction for the j th observation of the validation set.

RESULTS AND DISCUSSION

Data set 1

The data matrix was centred but not standardized, prior to the calibrations. The condition number ($\lambda_{\max}/\lambda_{\min}$) of the variance–covariance matrix $\mathbf{X}^T\mathbf{X}$ was 3.15×10^8 . This value exhibited the high degree of collinearity of the spectral variables.

The results, given in Table 1, showed the very common feature that all biased methods performed better than OLS. RPCR and RR led to the smallest CV errors. RPCR introduced 13 principal components, whereas RR used the 16 available principal components. The improvement in RPCR

Table 1. Cross-validation error of prediction for OLS, PCR, RR, RPCR and PLS regression methods. Values of model parameters are given in parenthesis

	PCR (c^*)	RR (k^*)	RPCR (c^*, k^*)	PLS (a^*)
OLS				
3.038	1.289 (10)	1.182 (1.7×10^{-5})	1.181 (13, 1.7×10^{-5})	1.200 (9)

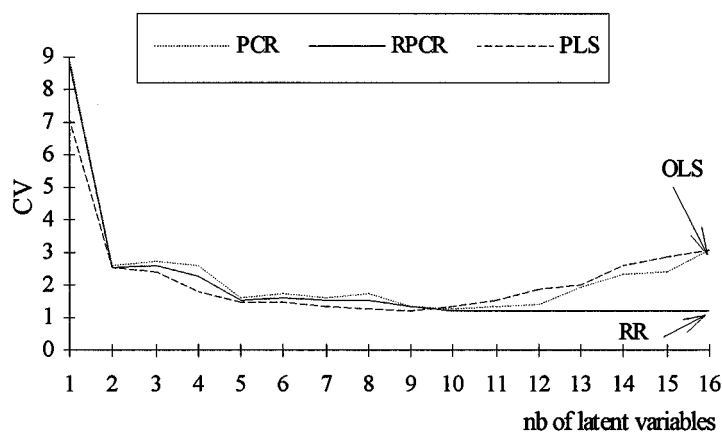


Figure 2. Data set 1. Plot of cross-validation error of prediction for five methods: PCR, OLS, RPCR, RR and PLS

over RR expected by the choice of a subset of principal components was not significant. On the other hand, RPCR, which amounts to adding a constant to a subset of eigenvalues, significantly improved the results of PCR. It can also be observed that the predictive ability of PLS was not better than those of RR and RPCR. This provides a case study where RR and RPCR led to a slight improvement over PCR and PLS, while Kowalski¹⁶ reported that the predictive performance of either PCR or PLS was better than that of RR on the basis of five data sets.

Figure 2 shows the behaviour of the CV criterion according to the number of latent variables introduced in the models. Regarding RPCR, a value k^* was determined for each value of c (number of principal components introduced in the model) by minimizing the PRESS criterion. A noteworthy feature of RPCR is the fact that the cross-validation error of prediction did not increase with the number of principal components. Unlike PCR and PLS, the performance of RPCR did not decrease by introducing higher degrees of collinearity. The RPCR solution reached its optimal level in so far as sufficient information was taken into account (here ten components) and the ridge additive constant k reduced the noise effect of the last components. This suggested that the first ten principal components contained useful predictive information and that the subsequent ones were associated with noise.

The comparison of the eigenvalues of $\mathbf{X}^T\mathbf{X}$ with the optimal values k^* of the different RPCR models for $c=1$ to p is given in Figure 3. This figure shows that the ridge parameter was of the same magnitude as the eigenvalues up to $c=10$, then k^* remained almost constant and greater than the last eigenvalues. The plateau observed in the plot of k^* was in correspondence with the plateau observed with the CV plot (Figure 2).

Data set 2

The collection considered in this second example refers to a common situation in NIR spectroscopy. Without prior selection of wavelengths, the data set had many more variables ($p=351$) than observations ($n=50$ for the calibration set). The condition number of the variance-covariance matrix $\mathbf{X}^T\mathbf{X}$ was 4.0×10^7 and exhibited the high degree of collinearity of the NIR spectral data. The first principal component, accounting for 99.2% of the total variance, was related to variations in the general intensity of the spectra. Moreover, a large number of the last eigenvalues were almost equal to zero (see Figure 6).

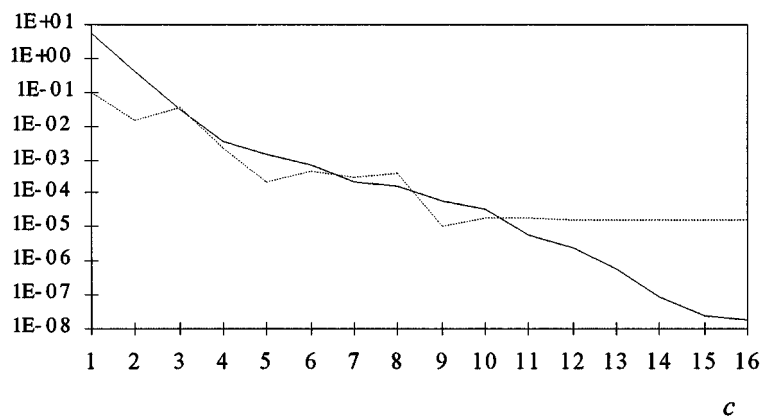


Figure 3. Data set 1. Plot of eigenvalues of $\mathbf{X}^T \mathbf{X}$ (full line) and plot of optimal selected ridge parameter in RPCR (dotted line) according to number c of principal components (logarithmic scale)

Both strategies were adopted in order to emphasize the discrepancy between the results derived from the LOO technique and $\text{PRESS}_{\text{formula}}$ in PCR. Figure 4 reveals the unreliable behaviour of $\text{PRESS}_{\text{formula}}$ when the number of components introduced in the model was large. Therefore the optimal parameter c^* in PCR was determined using the ordinary LOO technique.

Estimated cross-validated errors (equation (16)) are reported in Table 2 together with the optimal parameters selected for the biased regression methods. RPCR with twelve principal components and $k = 2.1 \times 10^{-6}$ performed better than the other methods considered herein. Nevertheless, the variations in performance of the four biased methods were relatively subtle in comparison with the OLS CV error.

In order to confirm the predictive ability of the methods, the fresh data of the validation set were included. The MSE_P criterion (equation (17)) was computed with the optimal parameters previously chosen on the basis of the calibration set. Results are given in Table 3. It appeared that RR and RPCR gave the smallest prediction errors for the validation samples. However, RPCR now performed slightly worse than RR, probably because the twelve principal components selected from the calibration set were not sufficient to describe all the spectral variations observed in the validation set.

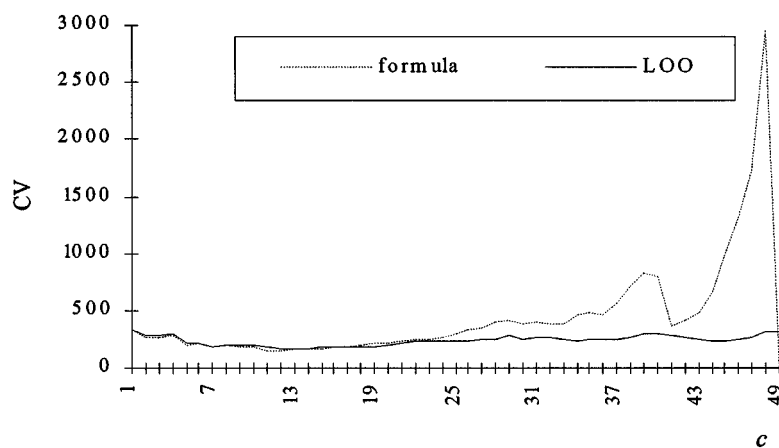


Figure 4. Data set 2. Evolution of CV criterion in PCR according to number c of principal components introduced in model, with two techniques of computation: LOO technique and 'formula' technique

Table 2. Cross-validation error of prediction for OLS, PCR, RR, RPCR and PLS regression methods. Values of optimal parameters obtained by $\text{PRESS}_{\text{formula}}$ or LOO technique are given in parentheses

OLS	PCR (c^*)	RR (k^*)	RPCR (c^*, k^*)	PLS (a^*)
320.94	163.90 (12)	161.95 (4.7×10^{-6})	153.76 (12, 2.1×10^{-6})	172.55 (7)

Besides the fact that the minimum errors of prediction were of the same order of magnitude for the four biased regression methods, the plot of MSEP in relation to the number of components introduced in the models (Figure 5) underlines an important feature of the methods. PLS and PCR were sensitive to the number of components in the models. The introduction of a large number of components may significantly damage the performance of these methods. In contrast, RPCR remained stable even if potentially noisy principal components were introduced. This property guards against any misspecification that might result from the introduction of a large number of components into PCR and PLS as a consequence of the presence of outliers in the calibration set.

Table 3. Mean square error of prediction for OLS, PCR, RR, RPCR and PLS using optimal values of parameters given in parentheses

OLS	PCR (c^*)	RR (k^*)	RPCR (c^*, k^*)	PLS (a^*)
255.87	155.59 (12)	138.11 (4.7×10^{-6})	144.96 (12, 2.1×10^{-6})	164.75 (7)

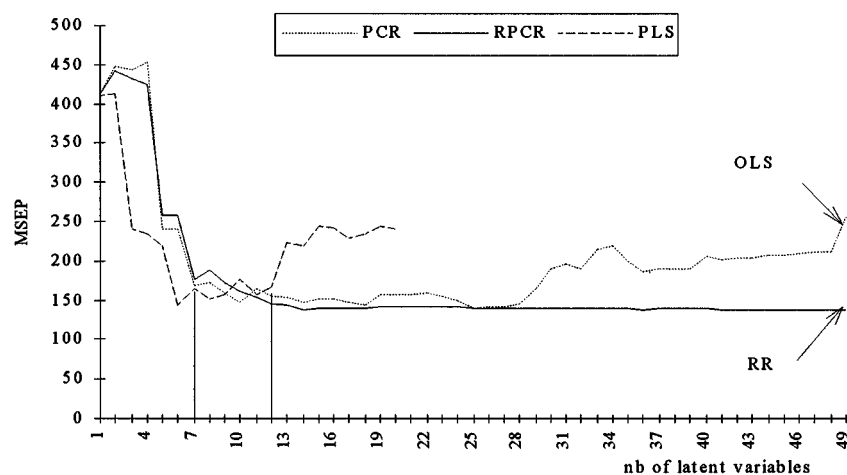


Figure 5. Data set 2. Plot of mean square error of prediction evaluated on validation set for five methods: PCR, OLS, RPCR, RR and PLS. Optimal values of parameters are indicated for PCR and PLS

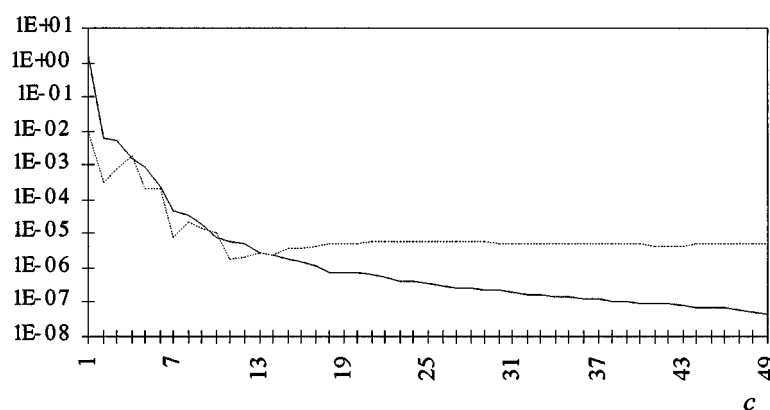


Figure 6. Data set 2. Plot of eigenvalues of $X^T X$ (full line) and plot of selected ridge parameter in RPCR (dotted line) according to number c of principal components (logarithmic scale)

Given this, RPCR may solve the awkward problem of determining the number of principal components encountered in PCR. A procedure that can be suggested is to remove those principal components unquestionably associated with noise and to add a ridge constant in order to counter the instability caused by the remaining small eigenvalues. For data set 2 a rational strategy might be to perform RPCR with 20 or 25 principal components in order to take into account all useful predictive information. The MSE_P for RPCR was 141.37 with 20 principal components and 139.55 with 25 principal components.

Figure 6 illustrates the comparison between the eigenvalues of $X^T X$ and the optimal parameters k^* for the RPCR models. As in the previous example (Figure 3), k^* decreased and then reached a plateau starting from $c=15$, which probably separates the signal from the noise as far as the prediction of y is concerned.

CONCLUSIONS

This study was a contribution to the use of biased regression methods, particularly PCR and RR, in spectroscopy calibration. A combination of these two well-known methods, called RPCR, was also introduced. Furthermore, PLS was performed as a complement to this. Illustrations were given on the basis of two data sets, the second one being a large NIR set collected in our laboratory.

Comparison was mainly based on the prediction ability of the methods. The first conclusion was that although RPCR or RR performed slightly better herein, the performances of the various strategies were similar. The behaviour of their prediction ability according to the number of components introduced into the models was another point of view. As parameters included in the biased estimation methods were determined by cross-validation using the calibration set, the expected stability of the performance is essential when a new data set is involved. Neither PLS nor PCR gave sufficient guarantees for this. Thanks to the addition of a constant k , the CV criterion as well as the MSE_P results associated with RPCR remained stable in so far as sufficient information was introduced into the model.

RPCR (and RR as a special case) copes with the awkward problem of choosing the number of components as it allows the inclusion of a large number of principal components (more than in the usual PCR). The inflation in the variance of coefficient estimates caused by those principal components with small variances is overcome by adding a ridge constant.

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